

Curriculum Learning for Inverse Scattering

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Abstract

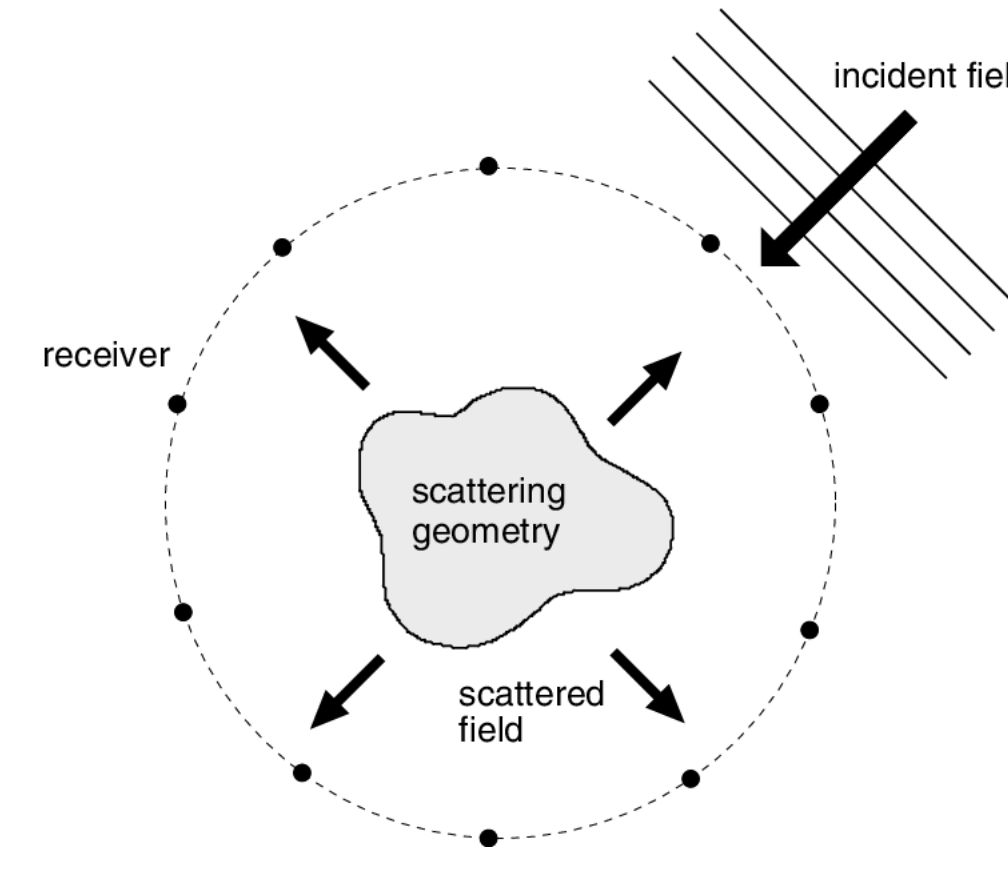
We address the inverse acoustic obstacle scattering problem in 2D for sound-soft obstacles and reconstruct the obstacle's boundary from scattered field measurements. Building on the neural network approach introduced in [4], which outperformed the classical linear sampling method (LSM) for initial domain estimation, we propose a curriculum learning strategy to enhance training. By progressively incorporating multifrequency data from low to high frequencies, our method improves accuracy, robustness to noise, and efficiency. Compared to [4], this approach yields faster, cheaper, more reliable training.

The Problem

The scattering process can be modeled by the following PDE:

$$\begin{cases} \Delta u + k^2 u = 0 & \text{on } \mathbb{R}^2 \setminus \bar{\Omega} \\ u = 0 & \text{on } \partial\Omega \\ \lim_{|x| \rightarrow \infty} |x|^{1/2} \left(\frac{\partial}{\partial |x|} - ik \right) u^{scat}(x) = 0 \end{cases}$$

$$u = u^{inc} + u^{scat}$$



Forward (Direct) Problem

The Forward Problem asks "Given a boundary $\partial\Omega$ and an incoming wave u^{inc} , what are the scattered waves u^{scat} ?"

We can use the Theory of Integral Equations and Potential Theory to solve the equation and get:

$$u^{scat}(x) = D(x) + i\eta S(x), \text{ on } x \in \mathbb{R}^2 \setminus \bar{\Omega} \text{ and } -u^{inc}(x) = [(\frac{1}{2}I + D + i\eta S)\varphi](x)$$

Inverse Problem

The Inverse Problem asks "Given the scattered field and incoming field u^{scat}, u^{inc} what was the domain Γ that created the scattered fields? In particular, we are looking for $\hat{\Gamma} = \partial\Omega$ such that

$$\hat{\Gamma} = \underset{\Gamma}{\operatorname{argmin}} \{ \|F(\Gamma) - u^{scat}\| \}$$

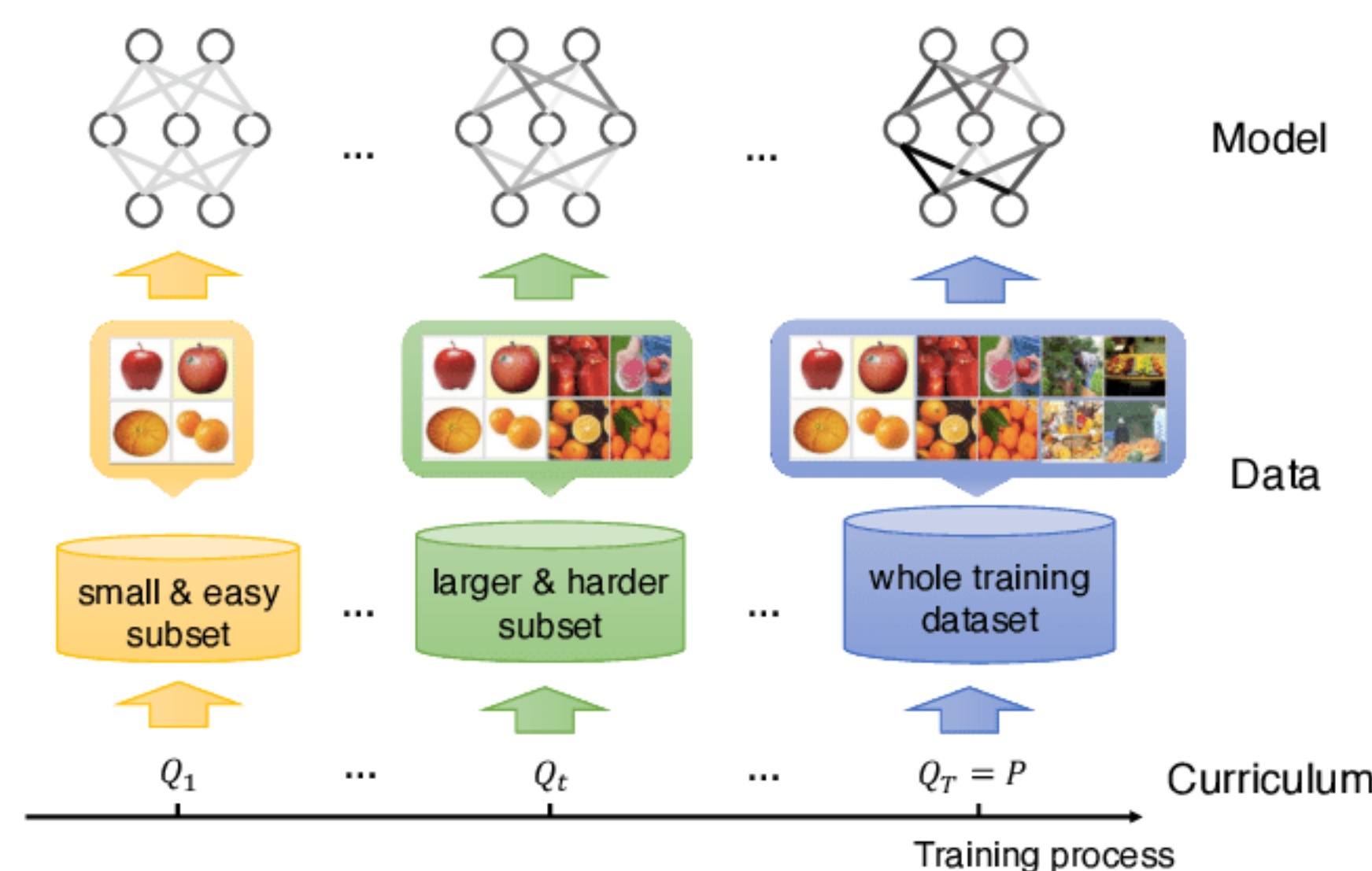
where $F(\Gamma)$ is the solution to the forward problem with boundary Γ .

Curriculum Learning

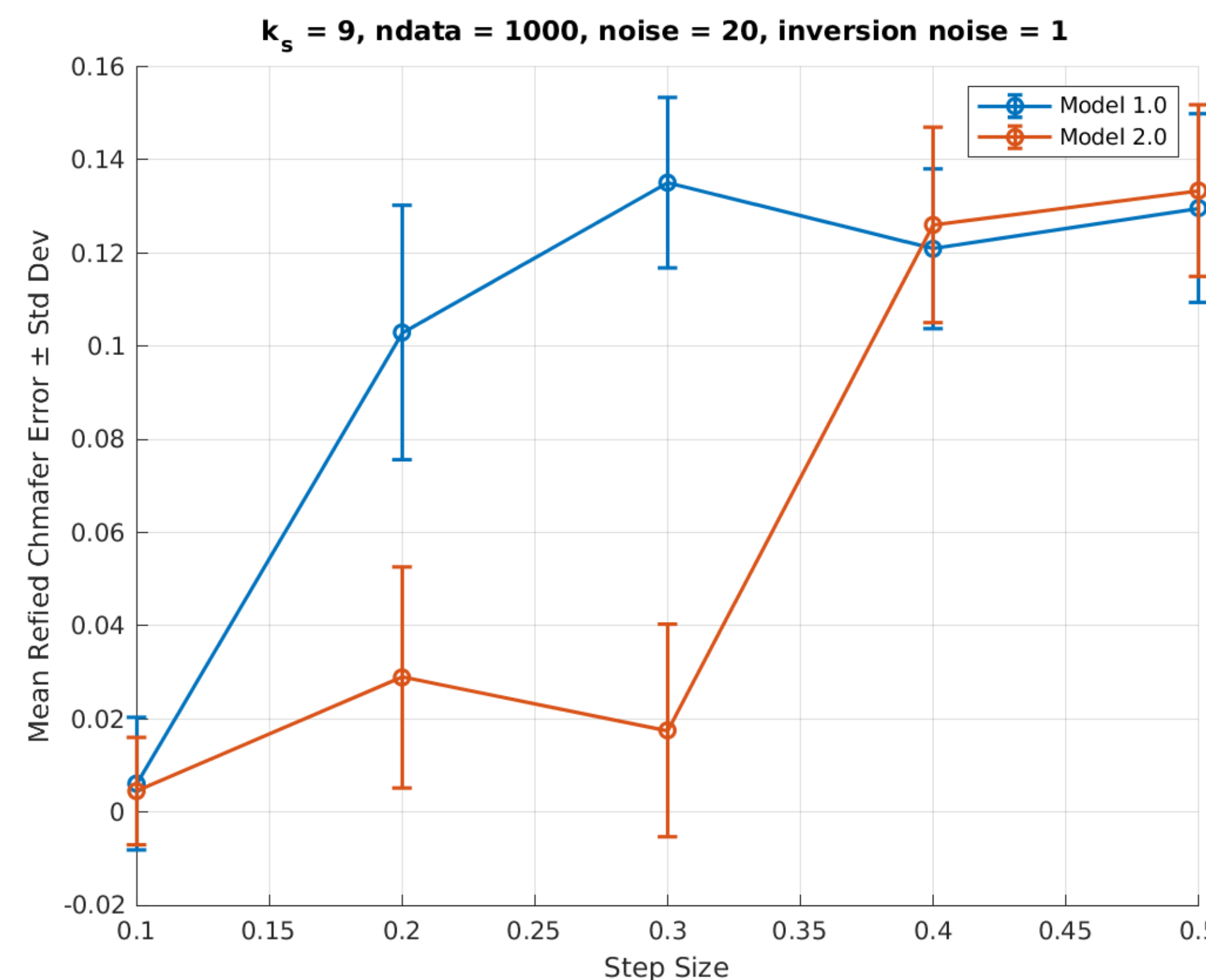
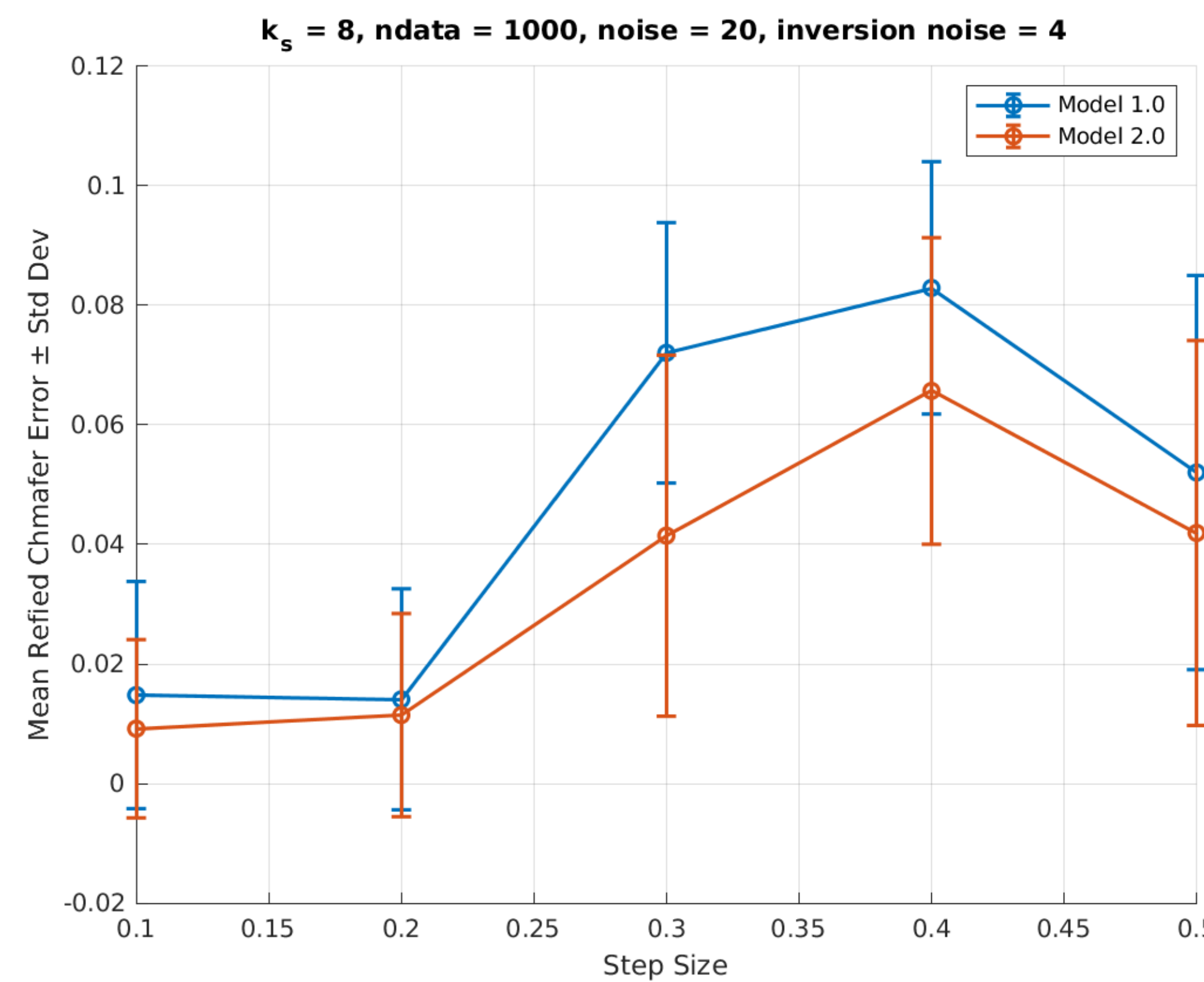
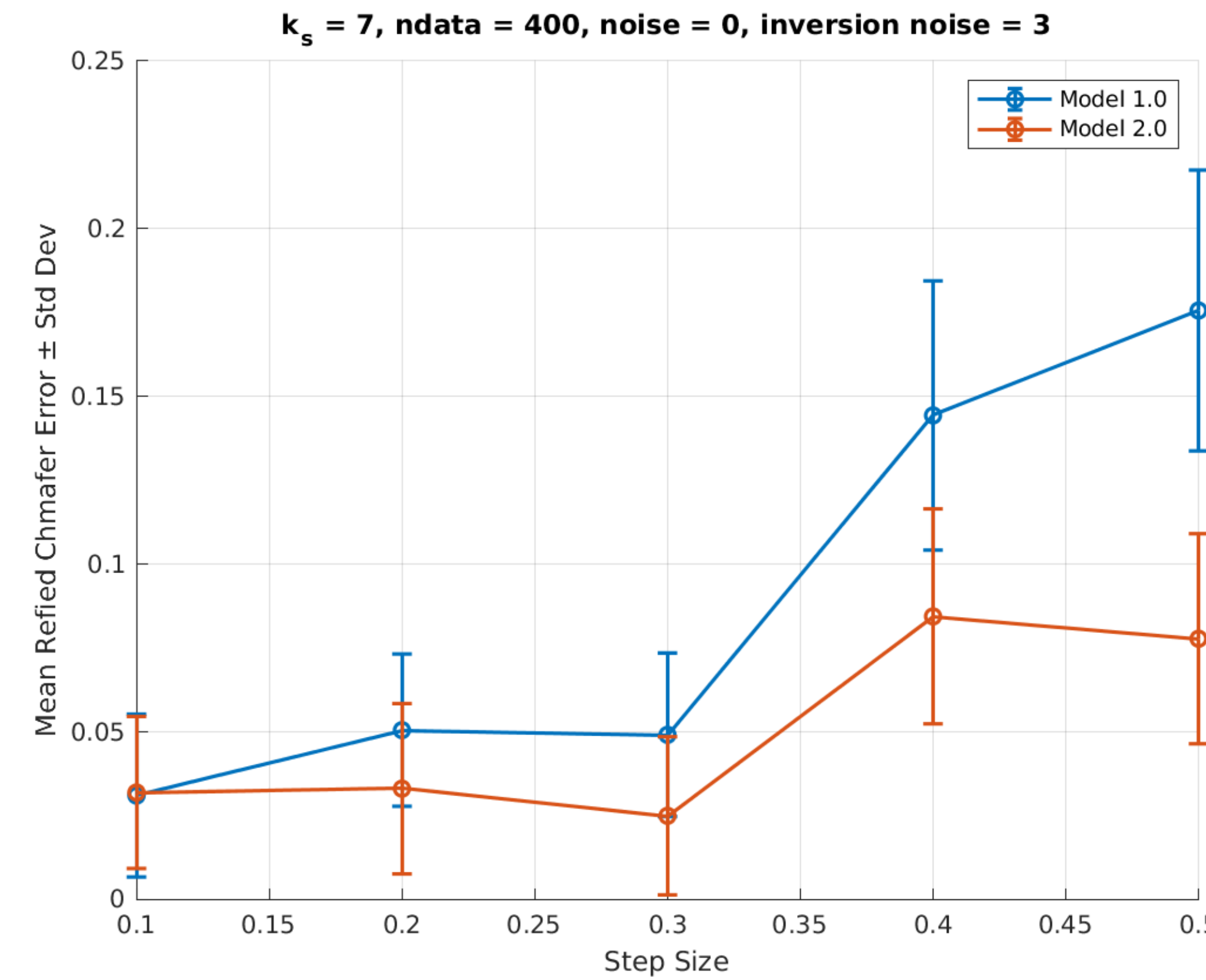
Motivation: Traditional iterative solvers for the inverse problem are accurate but can be slow or sensitive to initialization. ML can provide fast initial guesses, but training them on hard examples from the start can lead to poor convergence.

Curriculum Learning. Inspired by human learning, curriculum learning trains a network by gradually increasing the difficulty of the training data:

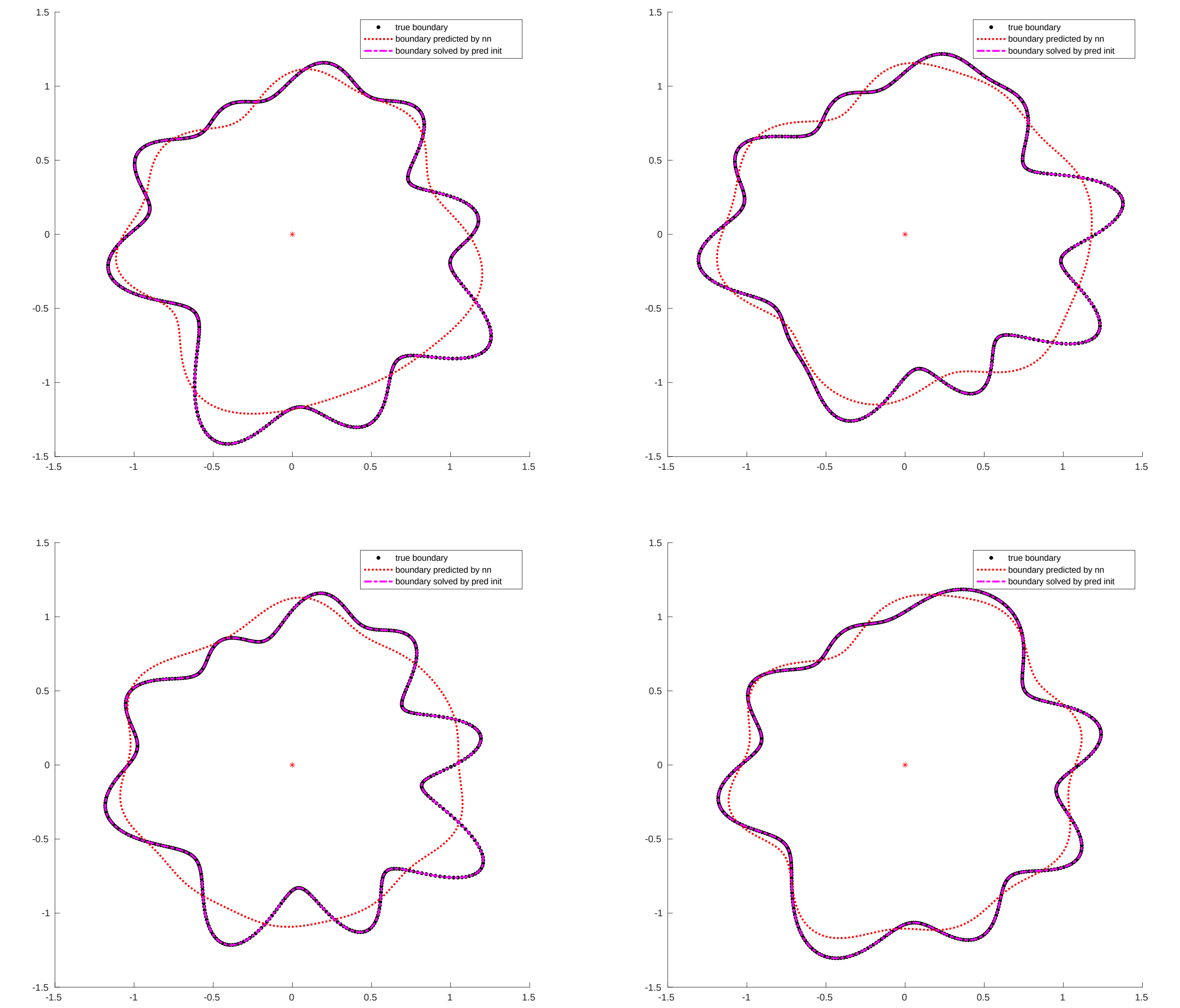
- Start with simple domains or low-noise data
- Progressively introduce more complex geometries and higher noise



How does Δk affect results?



Domains vs. The Model



The Process

- Generate Scattered Data**
 - Data Generation Parameters: ($k_{start}, \Delta k, k_{finish}, \text{noise}_{gen}$)
- Train a Neural Network to predict the domains**
 - Training Parameters: (N_{epoch}, N_{data})
 - We use a Convolutional Neural Network
- Use the Network's prediction as the 'Warm Start' for an Iterative Solver**
 - We use a Gauss-Newton like Solver that works by making small tweaks to the domain given the error in the previous guess

Research Computing Facts

Jobs Completed: $\approx 7,500$

CPU Hours Used: $\approx 90,000$

GPU Hours Used: $\approx 5,000$

Lines of Code: $\approx 16,000$

Amount of Data: $\approx 100\text{GB}$ over $\approx 1,000$ files datasets

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